

Module 3 Reflective Journal: Machine Learning Workflow

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Understanding the learning approaches

This lab helped me separate the three main machine learning approaches by asking what feedback the model receives. In supervised learning, the training examples include labels or correct answers. In the Wine dataset, chemical measurements were the input features and the wine cultivar was the target. Unsupervised learning instead searches for structure in data without known labels, such as grouping customers by similar behavior. Reinforcement learning is different because an agent interacts with an environment and learns from rewards or penalties. These approaches are not interchangeable; the problem and available feedback determine which one makes sense.

How the workflow fits together

The workflow made more sense when I viewed its stages as connected decisions rather than isolated tasks. First, the problem and data must be understood. Exploratory data analysis can reveal class imbalance, feature relationships, and possible quality issues. Then features and a target are selected, and the dataset is split into training and testing portions. Training teaches the algorithm from the training data, while evaluation checks predictions against examples the model did not train on. The confusion matrix adds detail by showing which classes were predicted correctly and where errors occurred. This is important because accuracy alone can hide the types of mistakes a model makes.

Model comparison and limitations

Comparing logistic regression with a decision tree showed that different algorithms can learn different decision patterns from the same data. A model that performs better on one test split is not automatically best for every future dataset. The chosen features, data split, model settings, and evaluation metric all affect the result. To improve confidence, I would use cross-validation to compare models, consider feature scaling for logistic regression, test other feature combinations, and examine precision and recall alongside accuracy. I would also keep the test set separate until the final evaluation to reduce the chance of tuning decisions to the test results.

Reflection and next questions

One broader lesson is that machine learning quality depends on the whole process, not just the algorithm. Data quality, appropriate validation, and careful interpretation are necessary before a model could be used responsibly. In an education example, a supervised model might identify students who could benefit from extra support using attendance and assignment patterns, but privacy, fairness, and human review would need to be considered. Questions I would like to explore further include how cross-validation changes confidence in a model comparison, how to identify the features most responsible for predictions, and how to monitor a deployed model when real-world data changes.

Personalize before submitting: Add one specific challenge you actually encountered, what you did to address it, and the actual model/accuracy reported by your executed notebook. This draft should reflect your own experience and course requirements.